

TUSCALOOSA, AL, USA

WMO: 722286

Lat: 33.212N

Lon: 87.616W

Elev: 46

StdP: 100.78

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93806

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.5	(c) -3.0	(d) -13.5	(e) 1.2	(f) -1.3	(g) -11.1	(h) 1.5	(i) 1.5	(j) 8.7	(k) 8.6	(l) 8.1	(m) 9.3	(n) 2.2	(o) 0	(p) 0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	36.3	24.1	34.8	24.3	33.7	24.1	26.5	31.9	25.9	31.4	25.5	31.0	3.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.4	28.4	24.5	19.6	28.0	24.0	19.0	27.7	82.6	31.9	80.3	31.6	78.2	31.0	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.6	6.8	5.7	-8.9	37.8	2.4	2.2	-10.6	39.4	-12.0	40.6	-13.3	41.9	-15.1	43.4		
			-9.7	27.6	2.4	0.9	-11.5	28.2	-12.9	28.7	-14.3	29.2	-16.1	29.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	7.3	9.8	13.8	18.0	22.4	26.3	27.6	27.5	24.5	18.5	12.5	9.1
	DBStd	8.30	5.71	5.45	4.96	3.89	3.33	2.13	1.69	1.92	3.16	4.48	4.71	5.20
	HDD10.0	322	123	65	22	1	0	0	0	0	0	1	27	83
	HDD18.3	1331	344	243	154	52	9	0	0	0	2	54	182	290
	CDD10.0	3301	39	60	139	242	386	488	546	542	436	264	103	54
	CDD18.3	1266	2	4	13	43	136	238	288	284	188	59	8	3
	CDH23.3	12442	3	20	127	427	1283	2392	2946	2887	1758	544	46	8
	CDH26.7	5526	0	1	17	90	482	1140	1433	1429	765	161	6	1
Wind	WSAvg	2.3	2.7	2.8	2.8	2.6	2.2	1.9	1.8	1.7	1.9	1.9	2.2	2.5
Precipitation	PrecAvg	1337	135	129	144	124	102	108	123	92	82	81	112	120
	PrecMax	1915	245	368	450	381	283	304	312	189	222	231	289	263
	PrecMin	900	28	33	30	10	11	9	22	19	20	3	25	25
	PrecStd	244	51	70	81	73	66	64	67	45	57	58	57	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.4	26.0	28.7	30.8	34.3	37.2	37.5	38.0	36.0	32.4	27.4	24.2
		MCWB	17.3	19.1	18.2	21.4	23.0	23.4	24.0	24.1	23.1	22.4	19.7	19.1
	2%	DB	21.3	23.1	26.5	28.9	32.3	35.2	35.9	36.3	34.0	30.2	24.5	22.1
		MCWB	16.9	17.4	17.7	20.0	22.4	23.8	24.8	24.3	23.3	21.1	18.0	18.1
	5%	DB	19.0	21.3	24.5	27.6	31.1	33.7	34.1	34.5	32.6	28.3	22.7	19.9
		MCWB	15.8	16.2	16.9	19.2	21.9	23.8	24.8	24.3	23.0	20.2	17.4	16.8
	10%	DB	17.1	19.0	22.5	26.2	29.4	32.4	33.0	32.9	31.2	26.7	21.1	17.8
		MCWB	13.8	14.7	15.6	18.3	21.4	23.6	24.7	24.4	22.7	19.7	16.7	14.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.6	20.5	20.8	22.9	25.3	26.6	27.3	27.1	25.7	24.8	22.3	20.9
		MCDB	22.1	23.6	24.6	27.9	30.5	32.5	32.2	32.5	31.1	28.6	24.1	22.7
	2%	WB	17.9	19.1	19.4	21.7	24.0	25.7	26.6	26.4	25.0	23.6	20.6	19.0
		MCDB	19.8	22.2	23.5	26.6	29.4	31.6	32.0	31.9	30.5	27.9	22.7	20.9
	5%	WB	16.4	17.6	18.3	20.7	23.2	25.2	26.1	25.8	24.4	22.5	19.1	17.5
		MCDB	18.9	20.1	22.6	25.2	28.4	31.0	31.6	31.2	29.6	26.0	21.6	19.3
	10%	WB	14.4	15.5	17.2	19.7	22.5	24.7	25.6	25.4	23.9	21.2	17.4	15.5
		MCDB	16.8	18.2	21.5	24.0	27.6	30.0	31.0	30.7	28.6	24.5	19.7	17.5
Mean Daily Temperature Range	5% DB	MDBR	11.5	12.0	12.8	13.3	12.2	11.5	10.9	11.1	11.9	13.0	13.0	11.3
		MCDBR	12.6	13.6	14.7	14.4	13.6	13.6	12.7	13.4	13.2	14.5	14.2	12.4
		MCWBR	8.9	8.4	7.5	6.7	5.1	4.4	4.0	3.9	4.4	6.3	8.2	8.6
	5% WB	MCDBR	10.5	10.9	11.8	11.6	10.9	11.0	10.8	10.6	10.5	10.2	10.2	10.2
		MCWBR	8.4	8.1	6.9	6.4	4.7	4.3	4.1	4.0	4.1	5.4	7.4	8.4
Clear-Sky Solar Irradiance	taub	0.324	0.335	0.365	0.401	0.450	0.482	0.533	0.532	0.443	0.373	0.347	0.324	
	taud	2.519	2.482	2.418	2.304	2.214	2.156	2.035	2.031	2.270	2.452	2.465	2.534	
	Ebn at Noon	893	918	912	888	844	813	769	763	822	863	856	873	
	Edh at Noon	84	97	111	130	144	152	171	169	128	99	88	79	
All-Sky Solar Radiation	RadAvg	2.65	3.18	4.34	5.50	6.06	6.29	6.04	5.55	4.91	4.07	3.01	2.27	
	RadStd	0.21	0.44	0.40	0.43	0.45	0.49	0.38	0.35	0.42	0.50	0.31	0.23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+143	+103

Nomenclature: See separate page